

# Aditya Shekhawat

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Portfolio

## PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years owning cloud-native, high-throughput services from design to production. Architected multi-agent LLM pipelines and async queue systems processing 10,000+ operations/day at 99.9% reliability at BrowserStack. Expertise in Java, Node.js, Python; hands-on with AWS, Kubernetes, Kafka, and the full SDLC.

## TECHNICAL SKILLS

|                 |                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------|
| Languages       | Java, Python, JavaScript/TypeScript, SQL, C++ (fundamentals)                                    |
| Backend         | Spring Boot (Security, WebFlux), Node.js (Express / NestJS), REST, GraphQL basics               |
| Databases       | PostgreSQL, MySQL, MongoDB, Redis (caching & pub-sub), DynamoDB, FAISS (vector)                 |
| Cloud & Infra   | AWS (S3, SQS, Lambda, CloudWatch, IAM), Docker, Kubernetes (Helm, k9s), Azure & GCP familiarity |
| Messaging       | Kafka, RabbitMQ, BullMQ — event-driven design, dead-letter queues, backpressure                 |
| AI / LLM        | OpenAI & Anthropic APIs, LangChain, RAG pipelines, embeddings, agentic orchestration            |
| Observability   | Prometheus, Grafana, OpenTelemetry, CloudWatch — dashboards, alerting, tracing                  |
| Testing & CI/CD | TDD, JUnit, Mockito, Jest, Integration Testing; Jenkins, GitHub Actions                         |
| Networking      | HTTP/S, TCP/IP, UDP, DNS, WebSockets — REST protocol design & debugging                         |

## EXPERIENCE

### Software Engineer I — Backend

Dec 2024 – Present

*BrowserStack*

*Remote, India*

- Architected and owned end-to-end cloud-native async pipeline service using BullMQ, processing 10,000+ test-case generation operations/day with 99.9% reliability and sub-500ms median latency; drove design decisions from PRD review through production rollout.
- Designed a multi-agent LLM orchestration system where planning, feature-extraction, scenario-generation, and deduplication agents collaborate autonomously — reducing manual test authoring time by ~60% and improving scenario accuracy by 40% vs. single-prompt baseline.
- Improved pipeline throughput by 45% through queue architecture redesign, robust dead-letter handling, and horizontal worker scaling; built Prometheus/Grafana dashboards and alerting that cut MTTR by 35%.
- Led integration of LLM-based feature extraction from PRDs into the TCM system; defined schema contracts, coordinated with PM and QA stakeholders, and shipped with zero production incidents.
- Enforced TDD (Jest unit + integration) across all pipeline modules and maintained GitHub Actions CI/CD with automated quality gates.
- Stack:** Node.js, BullMQ, WebSockets, Docker, OpenAI/Anthropic APIs, Prometheus, Grafana, GitHub Actions.

### Associate Software Engineer

Jun 2023 – Dec 2024

*Kanerika Software*

*Hyderabad, India*

- Owned design and delivery of ‘Phi’, an internal data-visualization product: architected the backend using Spring Boot REST APIs with PostgreSQL, handling real-time chart data for 50+ concurrent users with <200ms p95 response times; led full SDLC from requirements through deployment.
- Designed and enforced atomic-design component architecture across frontend (React/Redux) and backend services, reducing code redundancy by ~30% and cutting new-feature delivery time by 20%.
- Built and maintained GitHub Actions CI/CD pipelines with automated build, test, and deployment stages; achieved zero-downtime deployments across staging and production environments.
- Translated ambiguous stakeholder requirements into technical designs with trade-off analyses, reducing re-work by ~25% across 3 product cycles.
- Stack:** Java, Spring Boot, PostgreSQL, React.js, Redux, PrimeReact, GitHub Actions.

## PROJECTS

### AI Document Intelligence Platform

*Node.js, Java/Spring Boot, LangChain, FAISS, PostgreSQL, Docker, AWS S3*

- Architected a scalable RAG backend that ingests and indexes large document corpora; supports natural-language queries over 100k+ page datasets with <2s end-to-end latency using hybrid lexical + semantic retrieval (FAISS + BM25).
- Built high-velocity async ingestion pipeline with queue-based worker pool; sustained 500+ document uploads/hour without degradation, with dead-letter retry and CloudWatch alerting.
- Deployed on AWS (S3 + ECS) with Docker; instrumented with OpenTelemetry for distributed tracing; maintained full TDD discipline (90%+ coverage) and automated CI/CD via GitHub Actions.

### Distributed Rate-Limiter Library

*Java, Redis (Lua scripting), Spring Boot, Docker*

- Implemented a plug-in Spring Boot rate-limiter library using Redis atomic Lua scripts for token-bucket and sliding-window algorithms; handles 50,000 RPS with <1ms overhead per request.
- Published as an internal library consumed by 3 microservices; includes configurable per-user/IP/endpoint limits, real-time Grafana dashboard, and graceful degradation under Redis failure.

## ACHIEVEMENTS & RECOGNITION

- 4-Star Coder @ CodeChef** — Peak rating 1829; 300+ contest problems solved across long, cook-off, and lunchtime contests.
- CodeChef SnackDown 2021** — Qualified through to Round 1A of the 2nd phase of CodeChef’s flagship global programming championship.
- LeetCode DSA Solver** — Earned 100 Days, November 2021, and Algorithm I badges through sustained daily practice.

## EDUCATION

MBM University, Jodhpur — B.Tech, Computer Science | CGPA: 8.5 / 10

2019 – 2023